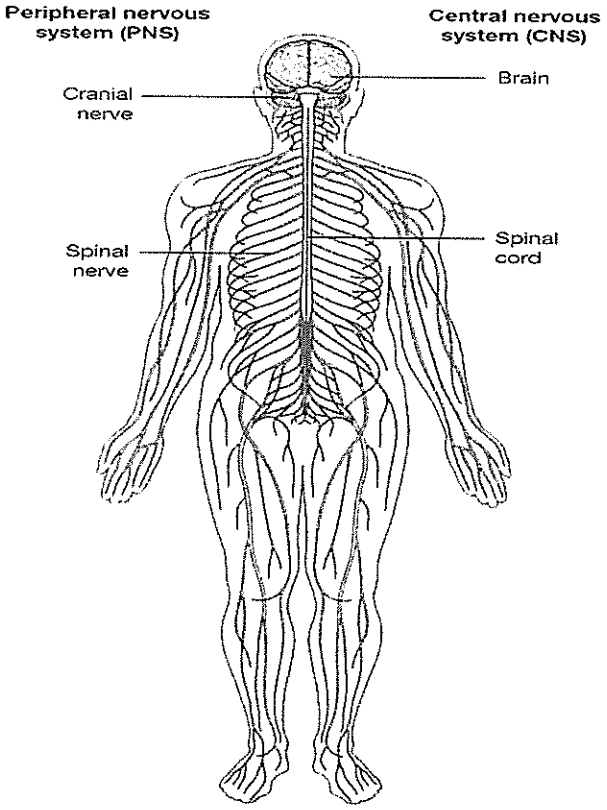
Getting on your Nerves???

The Nervous System is the master control system of the body and consists of the CENTRAL NERVOUS SYSTEM (CNS) and the PERIPHERAL NERVOUS SYSTEM (PNS) The CNS consists of the brain and the spinal cord and the PNS consists of 12 pairs of cranial nerves and 31 pairs of spinal nerves.

It regulates and controls all the body's functions through a complex system of nerves. The BRAIN is the central nerve centre that controls all thoughts and almost every movement. Messages from the brain to the various parts of the body are transmitted through NERVES



Drugs have varying effects on our Nervous System, in the way the nerve impulse is transmitted.

1 What would be the effect of using STIMULANTS and DEPRESSANTS.

_____

_____

2 Give an example of a drug that falls into these 2 categories.

_____

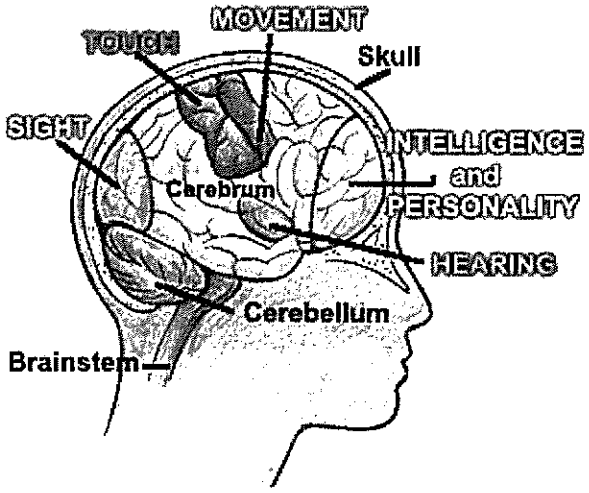
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The Brain

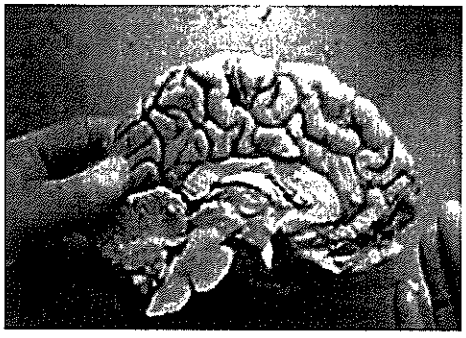
The brain is like a computer, with the senses providing the input, the brain processing the information and the instructions being sent to the muscles and glands are the output. The brain is a very delicate and vital organ of the body, which needs to be protected from injury, with the main protective structure being the cranium.

The cerebrum is the biggest part of the brain, and it is involved in mental activities such as thinking, reasoning, learning and memory.

The cerebellum lies underneath the cerebrum and is the second largest part of the brain. Its major function is controlling posture and balance, and the fine coordination of voluntary muscle movement.



The diagram below shows the dissection of a sheep's brain. Notice the cerebrum is the squiggly section across the top and the cerebellum is underneath the thumb of the person holding it.



What structures protect the brain under the skin & skull?

_____

THE BRAIN

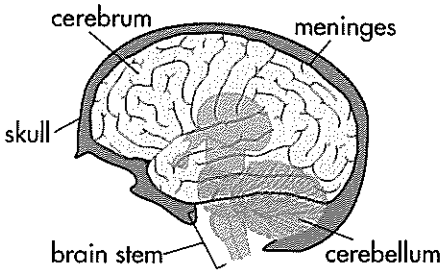
In this worksheet you will learn about how different parts of the brain have different jobs.

Our brain controls our actions, movements, thoughts and feelings. The brain weighs about 1.3 kilograms and is made up of millions and millions of neurones. You learnt about neurones in worksheet 13.1. The brain has different parts, each with specific jobs.

special words

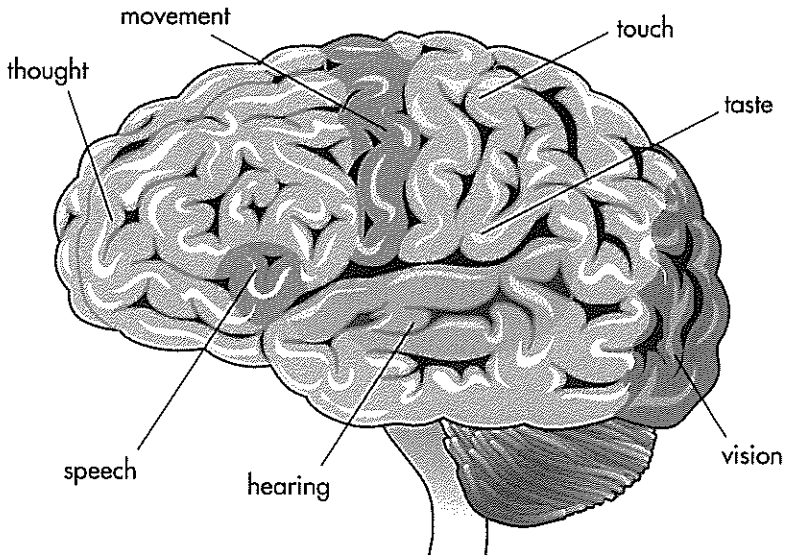
brain
neurone
skull
meninges
meningitis
cerebrum
cerebral hemisphere
cerebellum
brain stem

The skull is the thick bone that protects our brain.



Parts of the brain

- Meninges: thin layers that protect the brain from infection. The very serious disease called meningitis is a bacterial or viral infection of the meninges.
- Cerebrum: makes up 80% of the brain. It has two halves called cerebral hemispheres. These control our conscious thoughts and actions. The right hemisphere controls the left side of the body and the left hemisphere controls the right side of the body. Damage to one side of the brain may result in loss of control to the opposite side of the body. Different parts of the cerebrum control different functions. Look at the diagram below.
- Cerebellum: controls our balance and coordination.
- Brain stem: controls breathing, blood pressure and heart rate.



Functions of the cerebrum.

To do



1 Complete the table using information on this worksheet.

Part	Job
Brain stem	
	makes up 80% of the brain, different parts control different functions
Skull	
	controls our balance and coordination
Meninges	

2 Challenge! This is a research activity and could be done at home. Doctors can replace damaged hearts, lungs and livers using transplant organs. Transplant organs come from other people who have been declared dead.

Do you think doctors should be able to use a brain transplant to replace the brain of someone who is born with a serious brain defect? What are some of the arguments for and against doing a brain transplant?
Write an explanation of your choice in your workbook.

3 Share your ideas with a partner. Did you agree or disagree?

4 Do you think brain transplants are likely to take place in future? Explain your answer.

THE NERVOUS SYSTEM

In this worksheet you will learn how the nervous system controls the body.

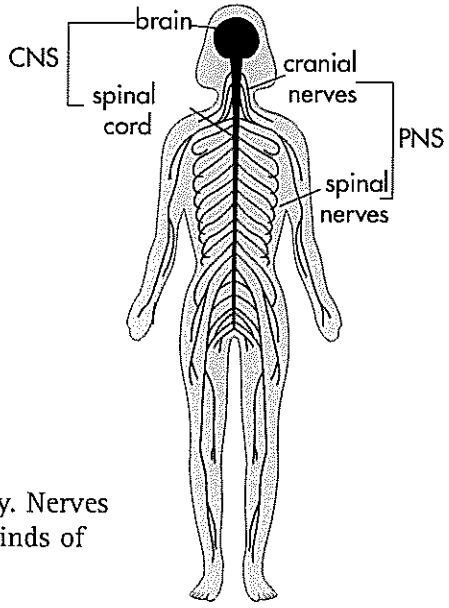
special words

nervous system
central nervous system (CNS)
brain
spinal cord
peripheral nervous system (PNS)
nerve cell
neurone
myelin
neurotransmitter
sensory neurone
motor neurone

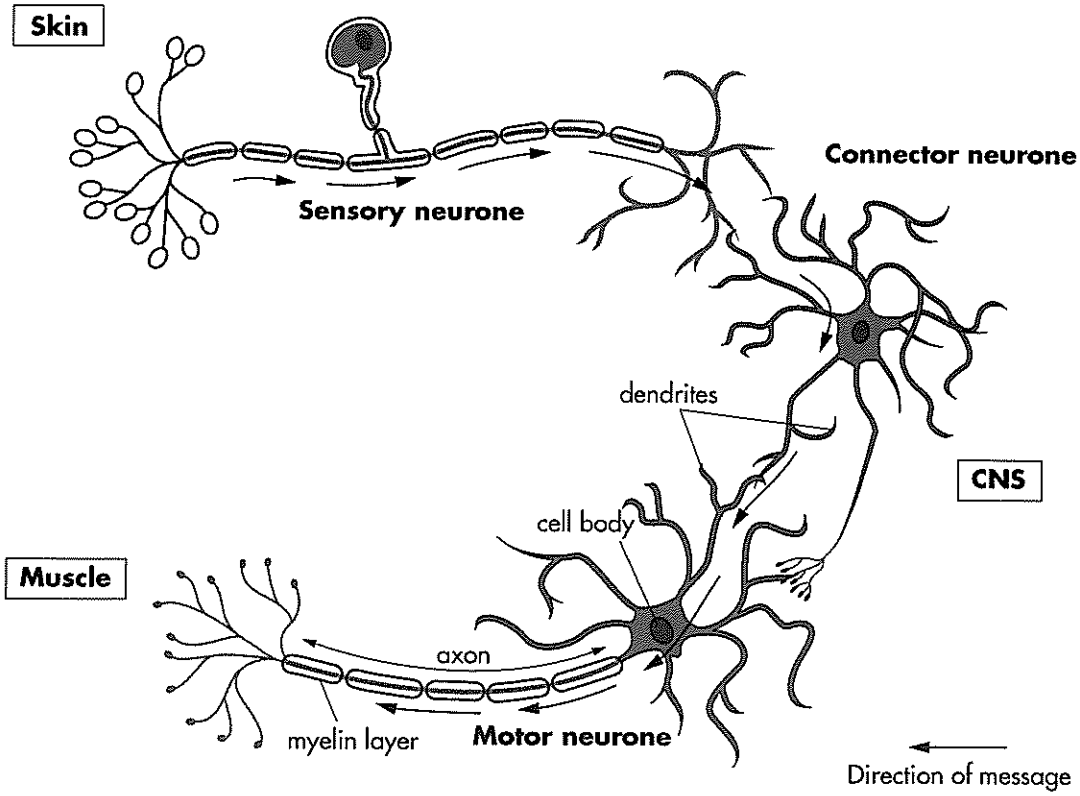
The nervous system is one of the control systems for our body. Our nervous system is spread throughout our body and works very quickly. It helps to coordinate various parts of the body and to steer us clear of dangers in our everyday lives.

The nervous system is made up of two parts:

- *central nervous system (CNS)* is made up of the brain and the spinal cord.
- *peripheral nervous system (PNS)* is made up of nerve cells (neurones) that connect all parts of the body to the brain and spinal cord.



Neurones carry messages to and from all parts of the body. Nerves are made up of bundles of neurones. There are different kinds of neurones; each has a different job and structure.



Messages only travel in one direction in neurones (nerve cells). Connector neurones pass the messages through the CNS.

Structure of neurones

- Neurones have a cell body with a nucleus.
- Dendrites are branching bits of the cell body that receive messages from other neurones.
- Axons carry the message along the neurone away from the cell body. The messages travel very fast, up to 100 metres per second. Messages only move in one direction.
- An insulating layer of myelin covers the axon of long neurones. Myelin helps the message to move quickly and stops it jumping between neurones. This stops the message from getting lost.
- Special chemicals called neurotransmitters pass the messages between neurones. Neurotransmitters are used up during activity and are replaced when we sleep. When we lack neurotransmitters we can become clumsy. This helps to explain why we sometimes bump into or drop things when we are tired.

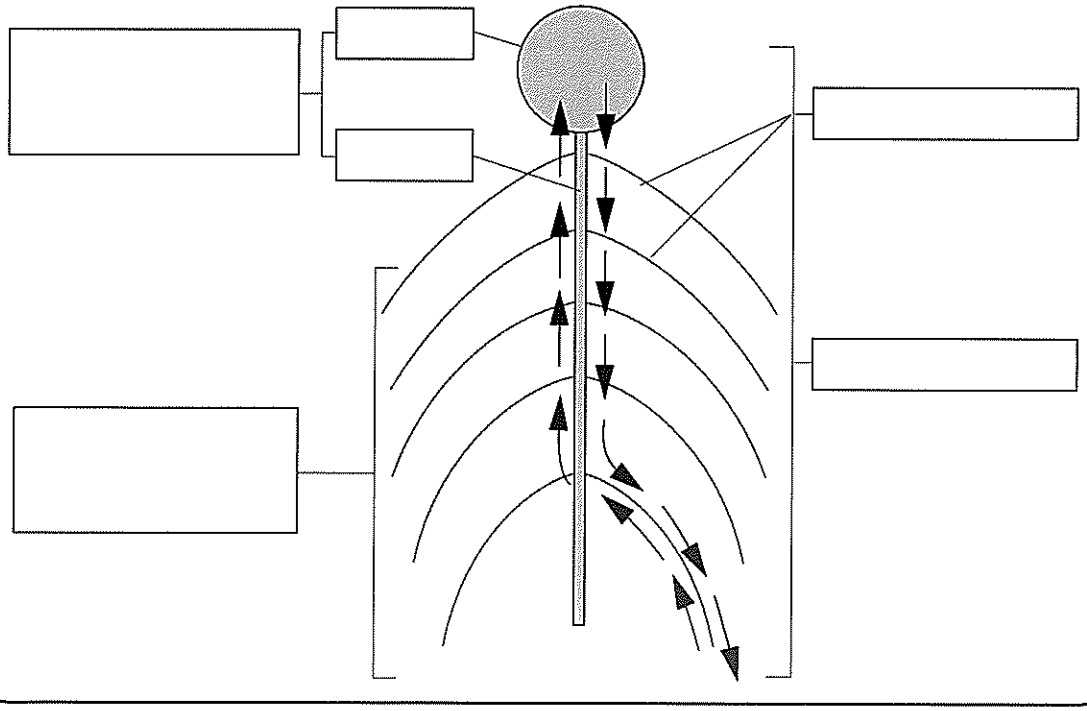
Messages pass from:

- the area where the message is received by sensory neurones
- to the spinal cord
- to the brain
- through the brain
- back down the spinal cord along motor neurones to an area causing a response.

To do

1 Use the words on this page to label the diagram.

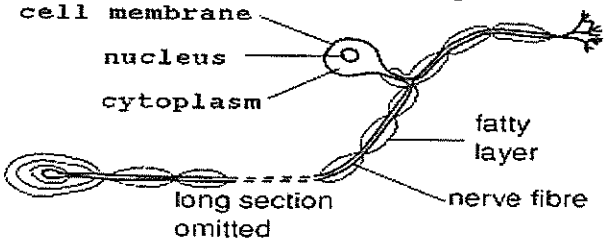
2 Your finished diagram will show the path messages take around our body.



Neurons....

The basic structural and functional unit of the CNS is called the NEURON, they vary in shape and size, but consist of similar features.
There are 3 different types of neurons, depending upon the function they perform:

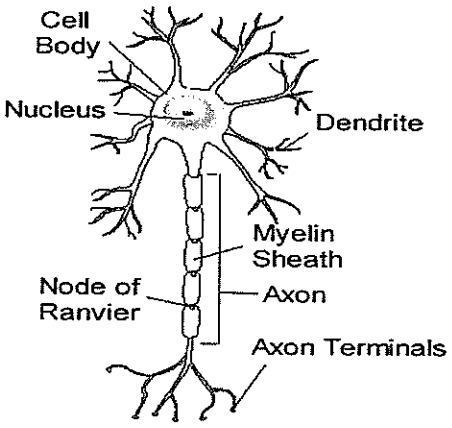
1. S _____Neurons- carry messages from the sense organs into the brain or spinal cord.



2. C _____Neurons- located inside the brain and spinal cord and are the link between sensory and motor neurons.



3. M _____ Neurons- carry messages from the brain and spinal cord to the effectors- muscles and glands.



4. What is the effect of Motor Neurone Disease on the body?

_____

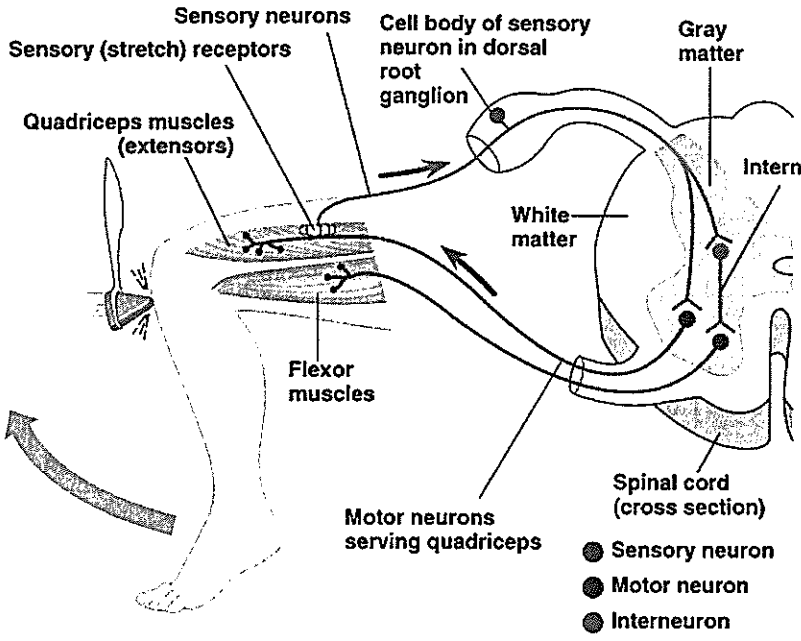
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What is a reflex??

A reflex reaction is something that your body does in response to a stimulus, such as a hit or a scare. You personally have no conscious control over the reaction. Like when someone hits your knee just under the knee cap and your leg bounces, or when you touch a hot stove and you instantly draw your hand away. It's a reaction that your brain doesn't have to think about, your body just does it automatically. This is called a reflex because it is not something you think about doing like picking up an apple, it's something your body does automatically, whether you think about it or not!

Simplest Nerve Circuit

- **Reflex, or automatic response**



- rapid response
  - automated
- signal only goes to spinal cord
  - no higher level processing
- adaptive value
  - essential actions
  - don't need to think or make decisions about
    - blinking
    - balance
    - pupil dilation
    - startle

The reflex arc consists of the 3 examples of neurons: the sensory, the connector and the motor neuron all working together to bring about this automated response.

List any other reflexes you have:

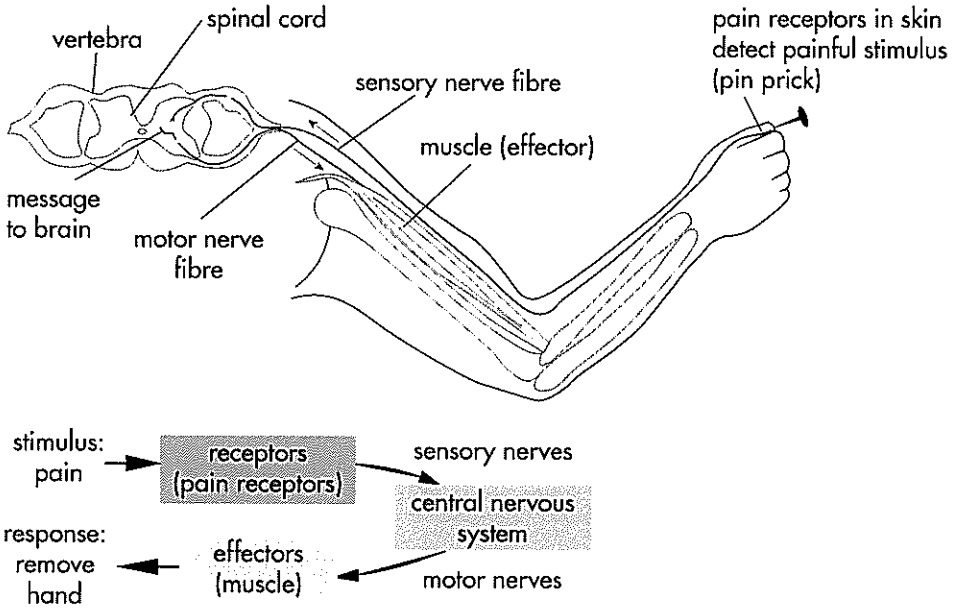
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_____

Some messages called reflexes are very fast.

- Reflexes are messages that travel from where they are received by sensory neurones to the spinal cord and then travel straight back along motor neurones to the same area to cause a response.
- Reflexes do not have to pass all the way up to the brain and back so they are very fast. They help protect us from dangers.
- Examples would be when you touch a hot object and pull your hand away, blinking your eyes to stop something getting into them, and pulling your foot away after standing on a sharp object.



The pathway of a reflex action can be represented by a reflex arc.

To do

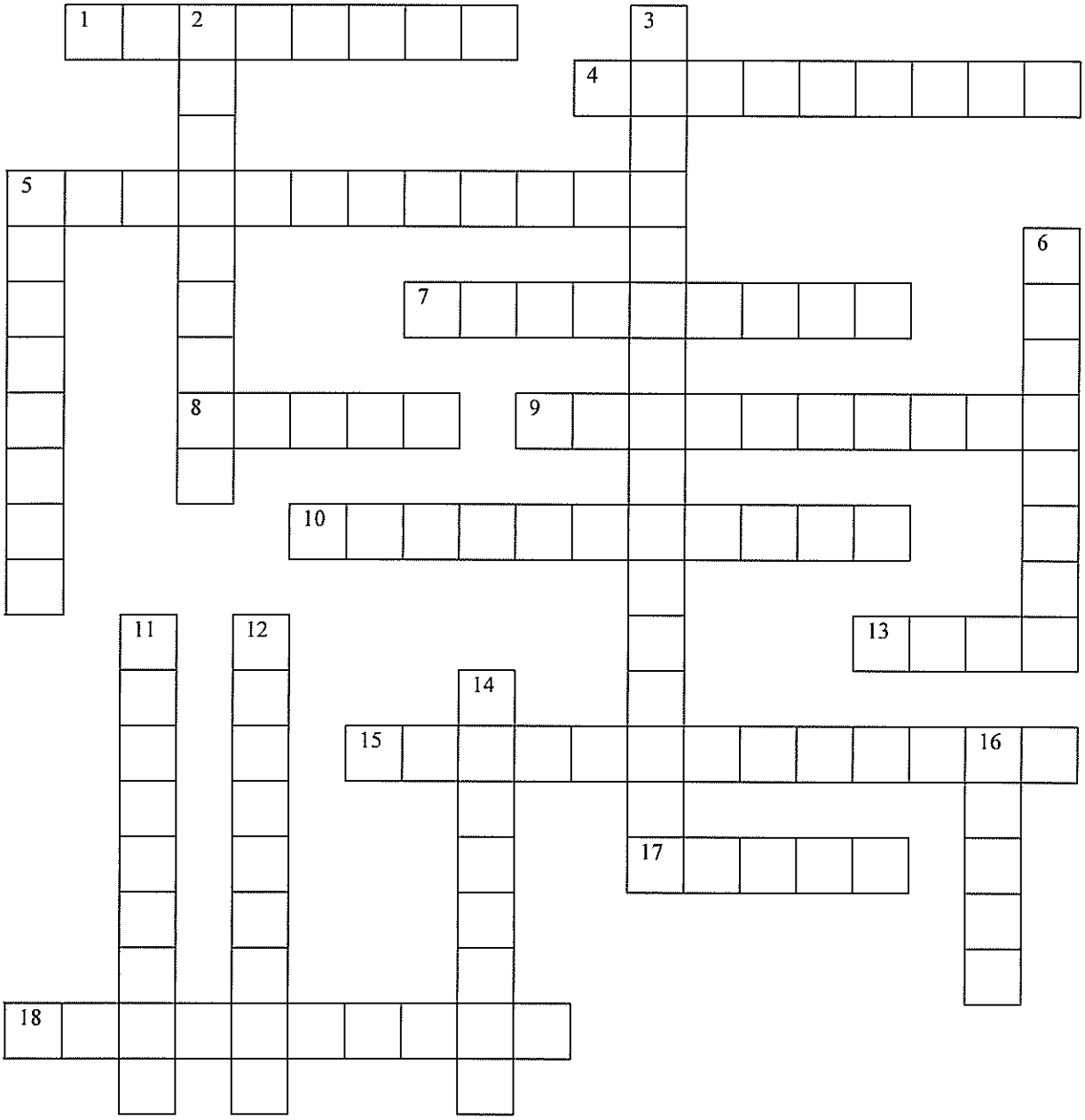


1 Use string, paper, Blu-tack, sticky tape and any other materials that you need to make a 2D model on a poster that shows how messages travel throughout our bodies. Your model should include sensory neurones, connector neurones, motor neurones, cell bodies, dendrites, axons and neurotransmitters.

2 Include arrows to show which way the message is travelling.

3 Explain your model to a partner.

Under control crossword



Across

1 The part that comprises 80% of the brain.

4 The hormone responsible for female characteristics.

5 The outer covering of nerve fibres. (two words)

7 The part of the nervous system that controls breathing, blood pressure and heart rate.

8 The creative and emotional side of the brain.

9 The part of the brain responsible for coordination and balance.

10 The outer protective covering of the eyeball.

13 The coloured part of the eye composed of muscle.

15 The jelly-like substance that fills the eye.

17 Thick bone protecting the brain.

18 The hormone that directs blood flow away from the digestive system to the brain and muscles.

Down

2 The pathway of a reflex action. (two words)

3 Receptors that are sensitive to mechanical stimulations such as touch.

5 Three thin membranes that cover and protect the brain.

6 Chemical messengers of the body.

11 The gland that controls other endocrine glands.

12 The type of gland that produces hormones.

14 The gas that causes fruit to ripen.

16 The type of nerve that sends messages from photoreceptors to the brain.

1 Tune Tune Tune Tune . 8	2 CCC PO	3 Reading
4 BRAIN	5 Me Quit	6 Ov Er
7 Step Far far	8 Endo	9 Straw Straw Straw Straw Straw Straw → Straw

10 Mula Mula Mula Mula	11 .GAP	12 R U feeling it it
13 kcab front front	14 oinut	15 morrow morrow never comes
16 Right Pla ce	17 man board	18 P O U R